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Issue/Rev.	Date	Section/Parag. affected	Reason/Initiation/Documents/Remarks
1	15/05/2026	all	First public release for 0.9

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1 Introduction

1.1 Scope

This document describes in details the `edps-gui` dashboard (Graphic User Interface), the recommended infrastructure to reduce data from the ESO telescope, using the ESO Data Processing System (EDPS). Each instrument has a dedicated tutorial that guides the user through the data reduction for that specific instrument.

1.2 What is EDPS?

The ESO Data Processing System (EDPS) is a framework to run ESO's data processing pipelines and it is meant to eventually replace the previous [ESOReflex environment](#). The general principles of EDPS have been described by [Freudling, Zampieri, Coccato et al. \[2024, A&A, 681, A93\]](#). Please refer to that paper if you have used EDPS for research resulting in a scientific publication.

Each of ESO's data processing pipeline consist of a series of standalone programs called *recipes*. Each recipe is designed to process certain type(s) of input data. The processing of these input data typically requires a range of auxiliary files such as calibration files. EDPS is designed to select appropriate input data for the different recipes of a pipeline, and execute them in sequence. This is done by specifying for each pipeline the workflow for organizing data and executing the recipes. This workflow can the used to process a set of data fully automatically.

1.3 Main concepts

EDPS is an environment designed to execute the recipes of an instrument pipeline according to a series of instructions. The main concepts in EDPS are:

- **Workflow and reduction cascades.** A workflow is a series of instructions designed to reduce data with an instrument pipeline in potentially multiple ways, by carrying on a sequence of tasks. Each workflow can define multiple reduction cascades, depending on the scientific needs. Foe example, the same workflow can be used to process data following different strategies that trigger different reduction steps (e.g. in one strategy flux calibration can be omitted) or different end-points (e.g., combine different science exposures, or stop after the reduction of individual exposures without combining them). Each of these "strategies" defines a "reduction cascade".
- **Task, jobs, and recipes.** A task is an element in the workflow that performs a given step of the data reduction cascade. Tasks are often associated to a recipe of the underlying instrument pipeline. A jobs is a work unit in a processing environment, that runs a recipe on a set of input data with a set of recipe parameters. A single task can generate several jobs: for example, a "bias" task, can generate multiple jobs, each of the running the bias recipe on a different set of input files.
- **Dataset.** A dataset is a collection of files, that are needed to perform the data reduction as specified by the workflow. It consists, for example, of one or more science files plus the calibrations needed to process them. In EDPS, the dataset has a hierarchical structure, which highlights the connections between the various files and tasks (e.g., task A is an input to task B).

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- **Target and Target category.** The "target", or the "target task" is the end point of the reduction cascade. When specifying a target, EDPS will process all and only the files needed to execute it. For example, if my target is "science", and the science files need the bias files, EDPS will process only the biases that have been selected to process those science files; then it processes the science using the product of the bias reduction. However, if my target is bias, then EDPS will process all and only the bias files, regardless they are not used by any science. In this case, EDPS does not processes the science, as it has reached already the end reduction point (e.g., process all biases). The "Target category" is a group of target that have similar purposes. For example, the target category "science", includes all the tasks that deliver final scientific products, the target category "qc1calib" includes all and only the tasks that processes calibrations (e.g., bias, standard stars).

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2 edps-gui: the EDPS dashboard

To install the EDPS dashboard (a.k.a. `edps-gui`), please follow the instructions given in the quick-guide [Section 2 \(installation\)](#).

To launch the dashboard, type:

```
. <path-to-environment>/bin/activate
edps-gui
```

The `<path-to-environment>` is the full-path-name of the virtual environment defined during the installation procedure.

The `edps-gui` is then loaded on a browser window (see [Figure 1](#)). To start EDPS, press the `Start EDPS` green button at the top of the window. To stop EDPS, press the `Stop EDPS` red button and press ‘Ctrl-c’ on the terminal where the `edps-gui` command was launched.

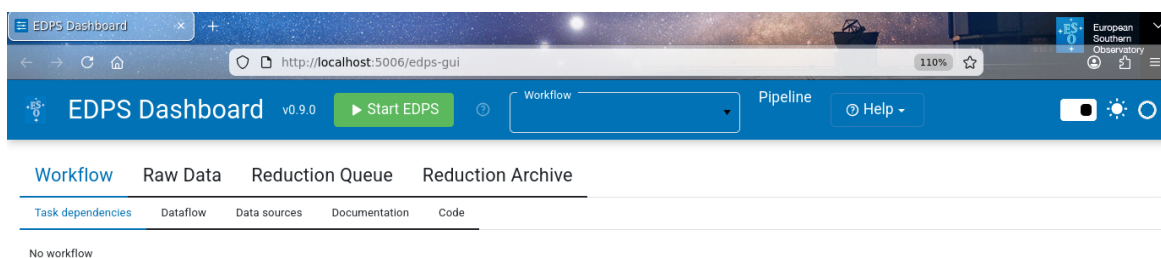


Figure 1: The `edps-gui` dashboard (Graphic User Interface). At this stage, EDPS has not been started yet and no workflow has been selected.

At the top of the Dashboard, near the `Start/Stop EDPS` buttons there is the **Workflow** selection menu, which allows to load the desired workflow. The available workflows depend on the pipelines installed in your system.

The `Help` drop-down menu contains version information, reference, and the **Settings** menu, which allows to configure several EDPS behaviors, such as links to data-storage directories, and the parallelization. It is described in detail in [Section 2.1](#). Note that it is possible to change the settings only if EDPS has not been started yet.

There are 4 tabs in the main dashboard window, they are:

- **Workflow.** It gives information on the workflow, the graphic layout, a description of the type of data and access to the workflow code. It is described in detail in [Section \[The **Workflow** tab\] \(#workflow\)](#).
- **Raw Data.** It allows to specify the input data, the preferences for association calibrations, and to create datasets to be reduced. It is described in [Section \[The **Raw Data** tab\] \(#raw_data\)](#).
- **Reduction Queue.** It starts the data reduction. It is described in [Section \[The **Reduction Queue** tab\] \(#reduction_queue\)](#).
- **Reduction Archive.** It gives access to previously archived data reduction and final products. It is described in [Section \[The **Reduction Archive** tab\] \(#reduction_archive\)](#).

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At the bottom of the dashboard, the graphic representation of the selected workflow is visible.

2.1 Settings